

Supplementary Information

Suppressing Internal Short Circuit Caused Thermal Runaway of Lithium-ion Batteries through Current Modulation

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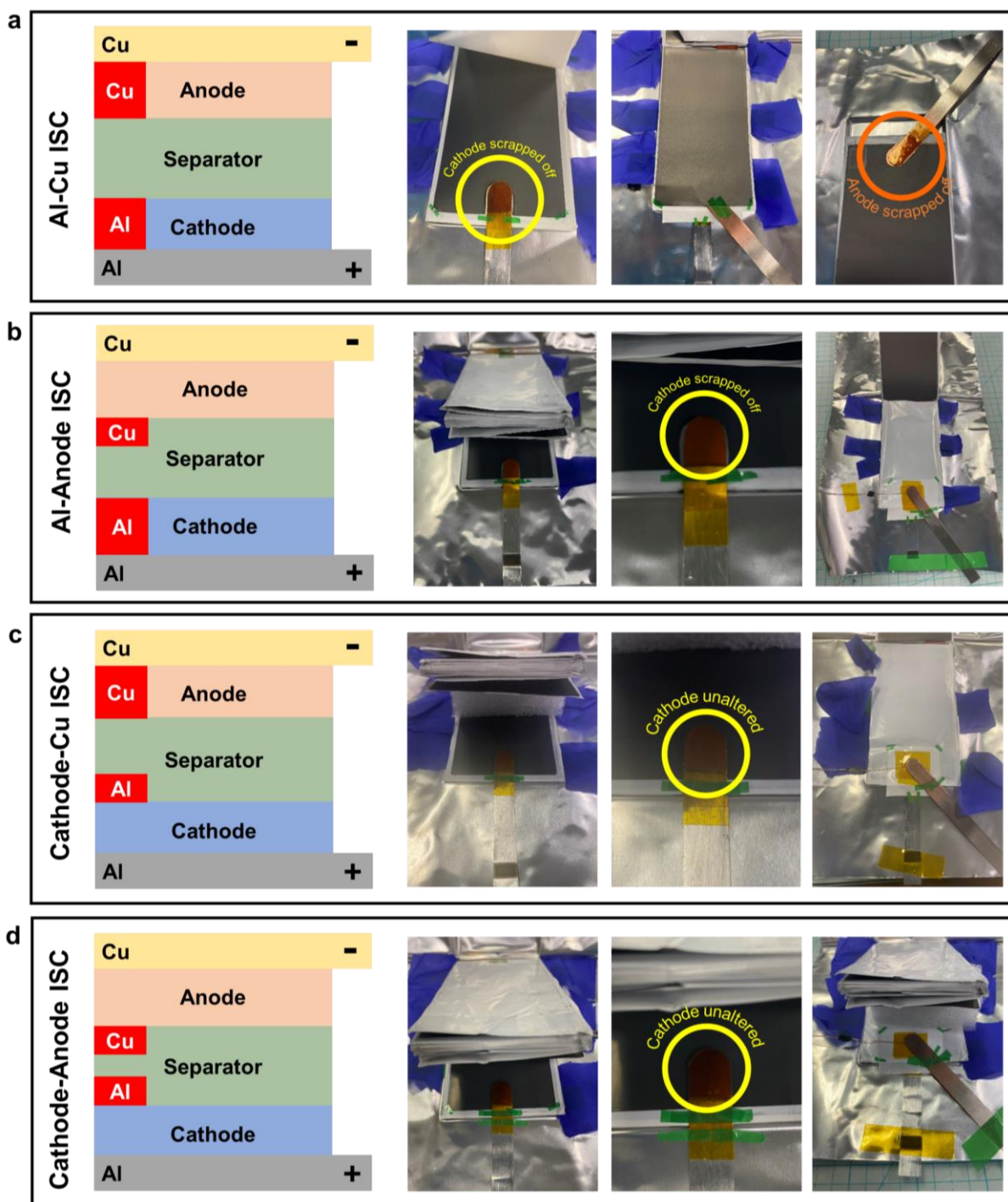
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List of Supplementary Figures

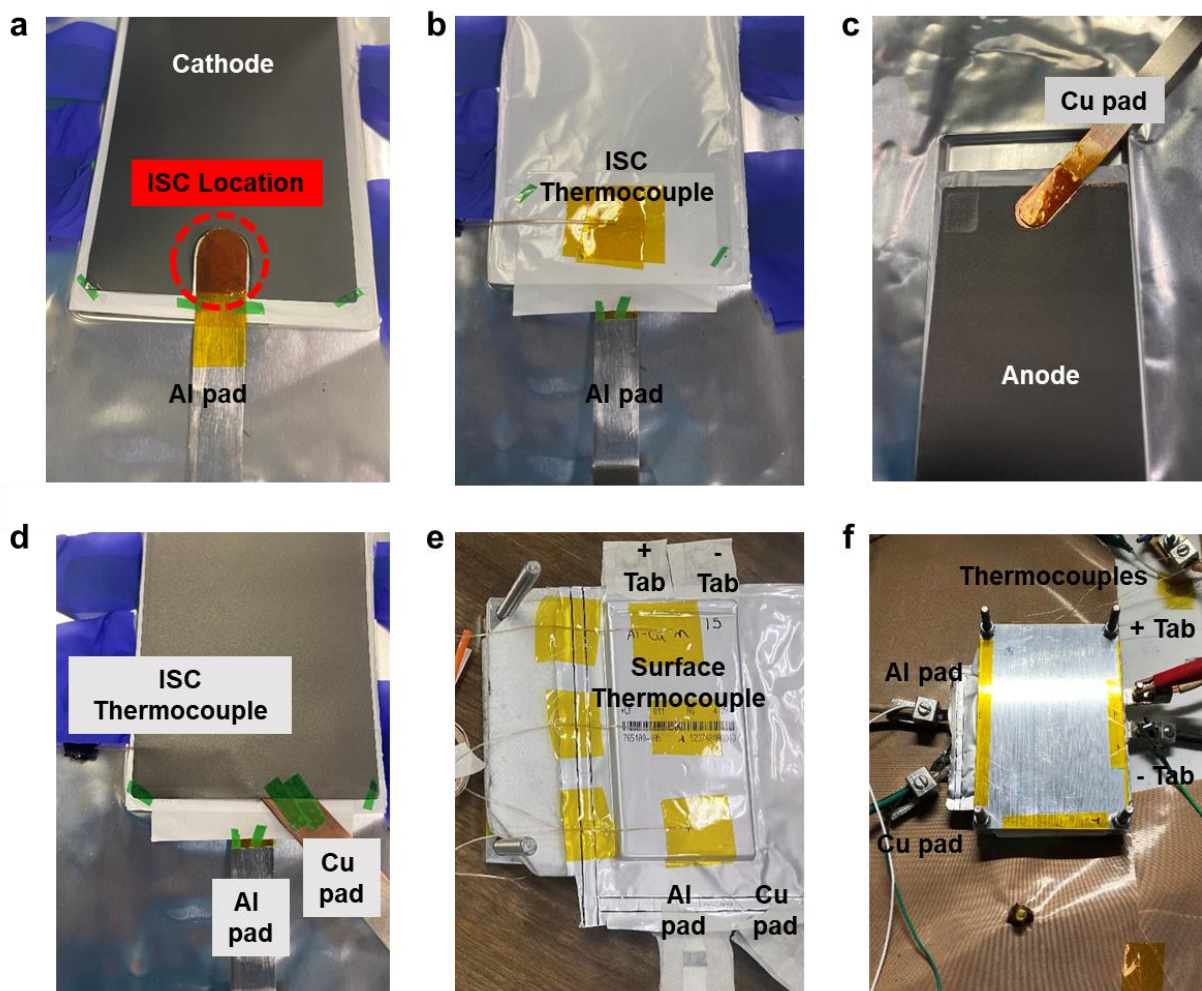
- Supplementary Fig. S1 | Creation of different types of ISC in Li-ion cells.
- Supplementary Fig. S2 | Process of fabricating Li-ion cells with Al-Cu ISC.
- Supplementary Fig. S3 | Process of fabricating Li-ion cells with Al-Anode ISC.
- Supplementary Fig. S4 | Discharge performance comparison of baseline cell and ISC cells.
- Supplementary Fig. S5 | Discharge performance of the Li-ion cell with Cathode-Anode ISC.
- Supplementary Fig. S6 | Discharge performance of the Li-ion cell with Cathode-Cu ISC.
- Supplementary Fig. S7 | Comparison of experimental results of Al-Cu ISC cells without and with current modulation.
- Supplementary Fig. S8 | Further analysis of cells with Al-Cu ISC and 120 A current modulation.
- Supplementary Fig. S9 | SEM images of separators facing anode without ceramic coating.
- Supplementary Fig. S10 | SEM images of separators facing cathode with ceramic coating.

List of Supplementary Tables

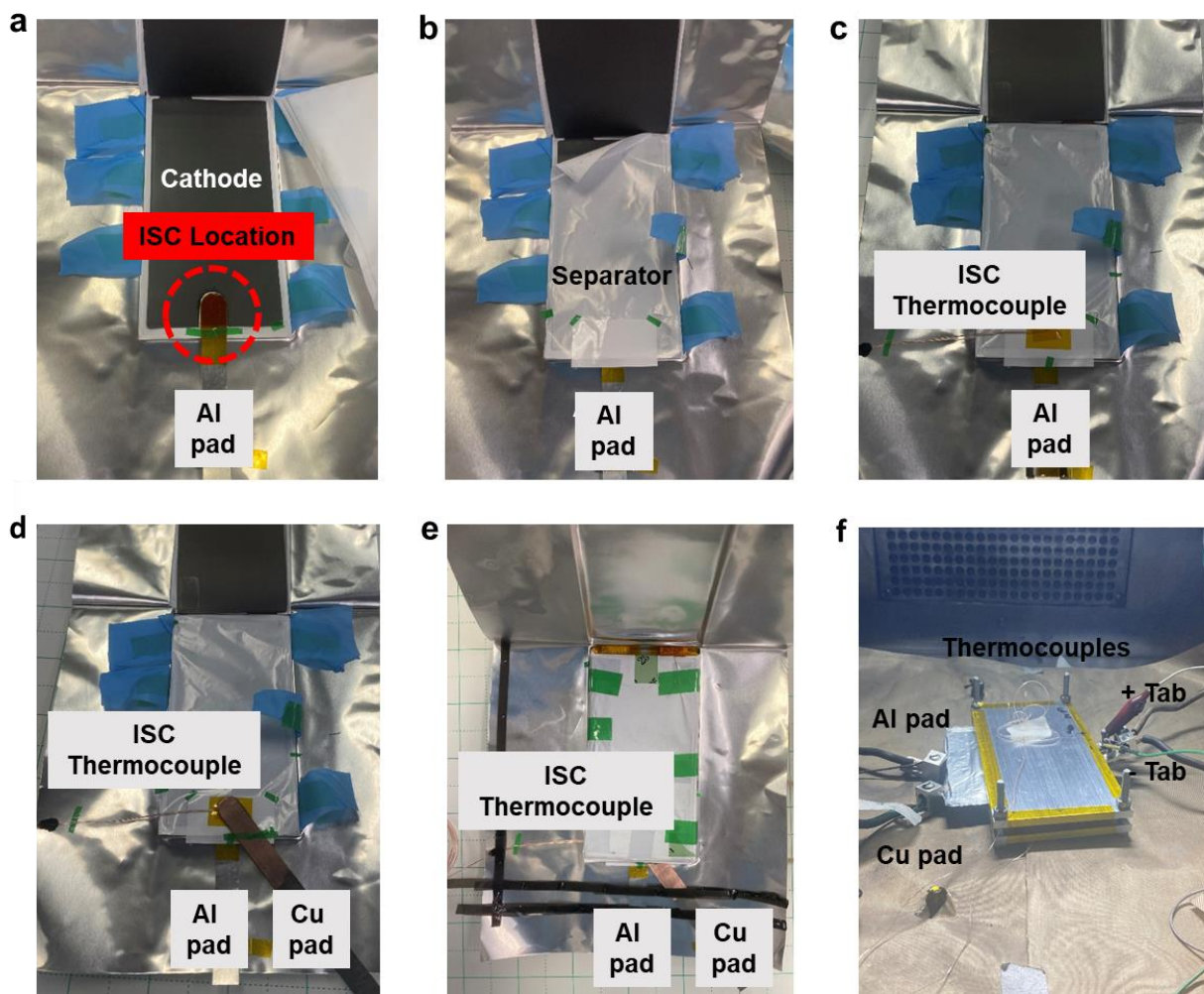
- Supplementary Table S1 | Energy balance analysis of Li-ion cells during Al-Anode ISC
- Supplementary Table S2 | Energy balance analysis of Li-ion cells during Al-Cu ISC without or with current modulation
- Supplementary Table S3 | Energy balance analysis Li-ion cells during Al-Anode ISC without or with current modulation
- Supplementary Table S4 | Charge capacity, discharge capacity, and coulombic efficiency of cells in formation cycles



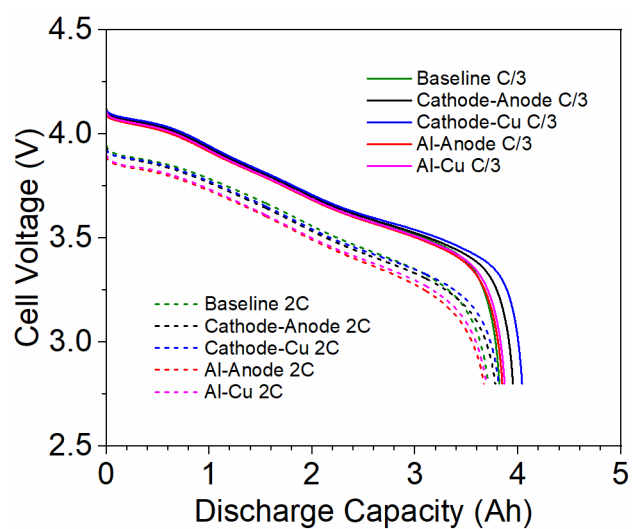
Supplementary Fig. S1 | Creation of different types of ISC in Li-ion cells. **a**, Al-Cu ISC: both anode and cathode are scrapped off at the ISC location, so the Al pad is in direct contact with the Al current collector and the Cu pad is in direct contact with the Cu current collector. **b**, Al-Anode ISC: only cathode is scrapped off at the ISC location, so the Al pad is in direct contact with the Al current collector. **c**, Cathode-Cu ISC: only anode is scrapped off at the ISC location, so the Cu pad is in direct contact with the Cu current collector. **d**, Cathode-Anode ISC: neither electrode is altered.



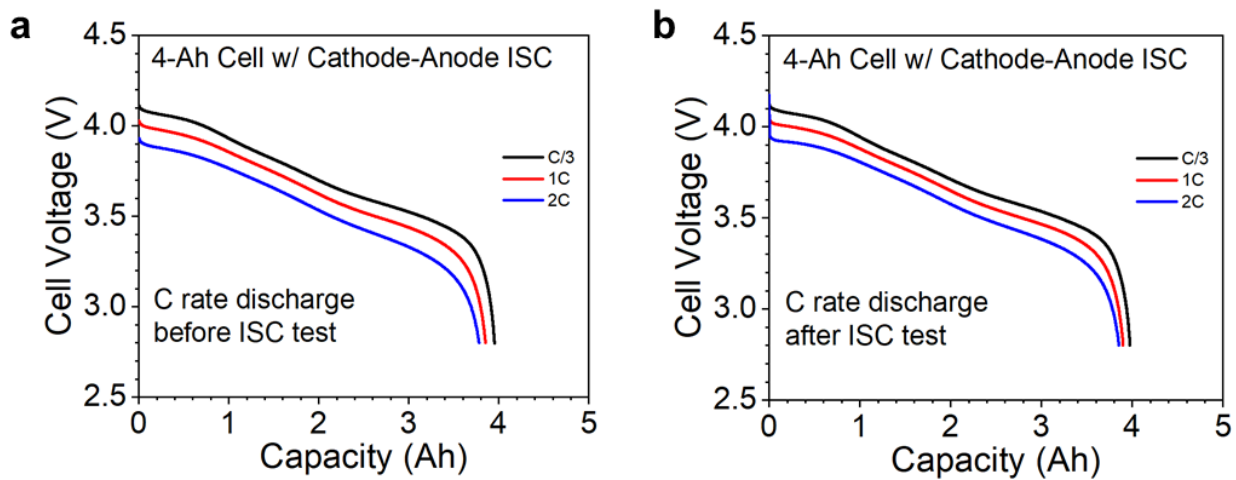
Supplementary Fig. S2 | Process of fabricating Li-ion cells with Al-Cu ISC. **a**, Placing Al pad directly onto the Al foil current collector after scrapping off the corresponding area of the cathode. **b**, Covering the Al pad with a separator before placing an additional piece of separator. **c**, Placing Cu pad directly onto the Cu foil current collector after scrapping off the corresponding area of the anode. **d**, Placing Cu pad above the thermocouple. **e**, Sealing the cell and placing surface thermocouples on the cell. **f**, A finished cell placed into the testing chamber.



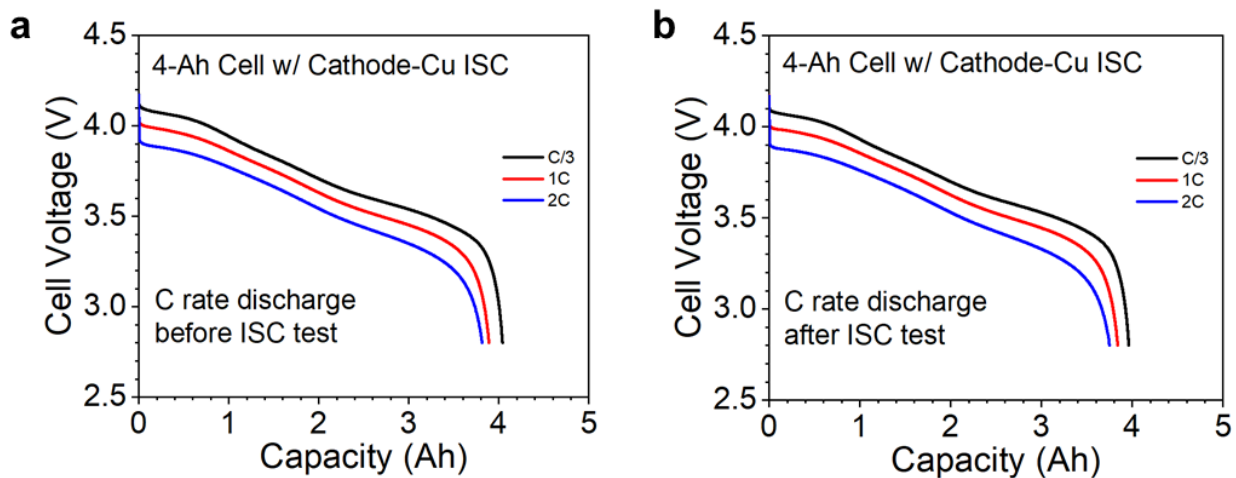
Supplementary Fig. S3 | Process of fabricating Li-ion cells with Al-Anode ISC. **a**, Placing Al pad directly onto the Al foil current collector after scrapping off the corresponding area of the cathode. **b**, Covering the Al pad with a separator before placing an additional piece of separator. **c**, Placing the ISC thermocouple on the added separator. **d**, Placing Cu pad above the thermocouple. **e**, Closing the cell and attaching heat shrink tape for sealing. **f**, A finished cell placed into the testing chamber.



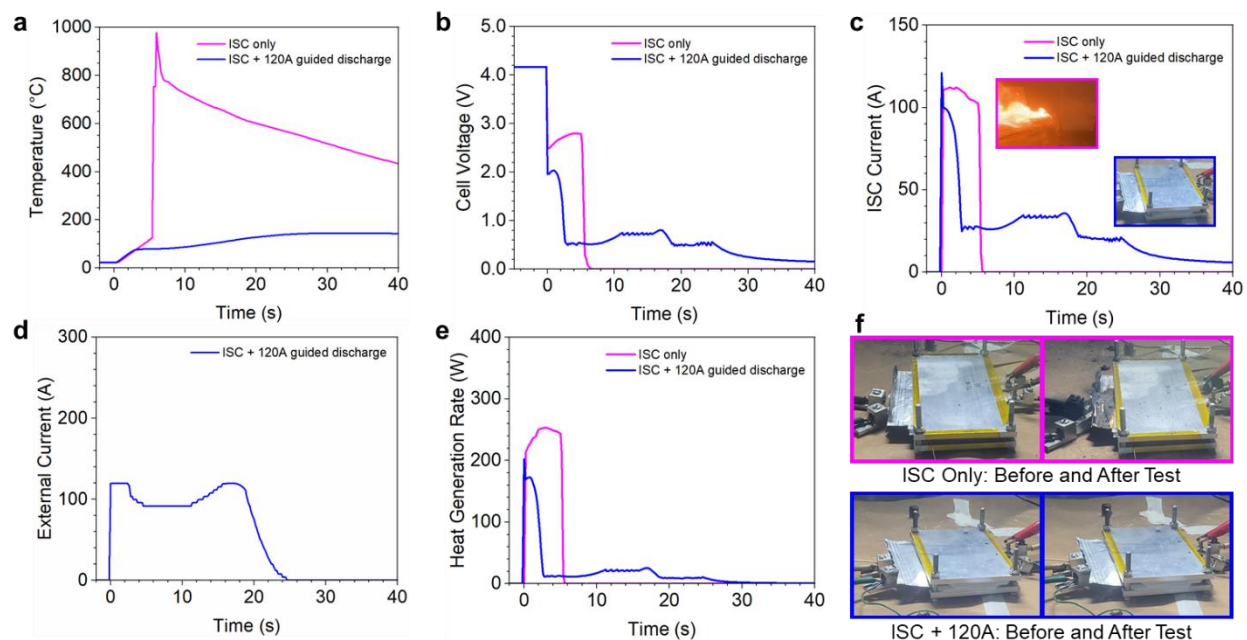
Supplementary Fig. S4 | Discharge performance comparison of baseline cell and ISC cells.
 The cells with different embedded ISC pads have similar performance to that of the baseline cell.



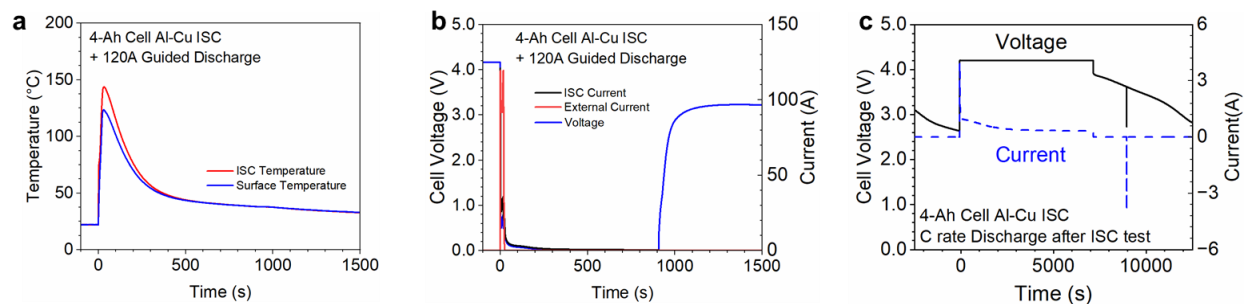
Supplementary Fig. S5 | Discharge performance of the Li-ion cell with Cathode-Anode ISC.
a, Before the ISC test. **b**, After the ISC test.



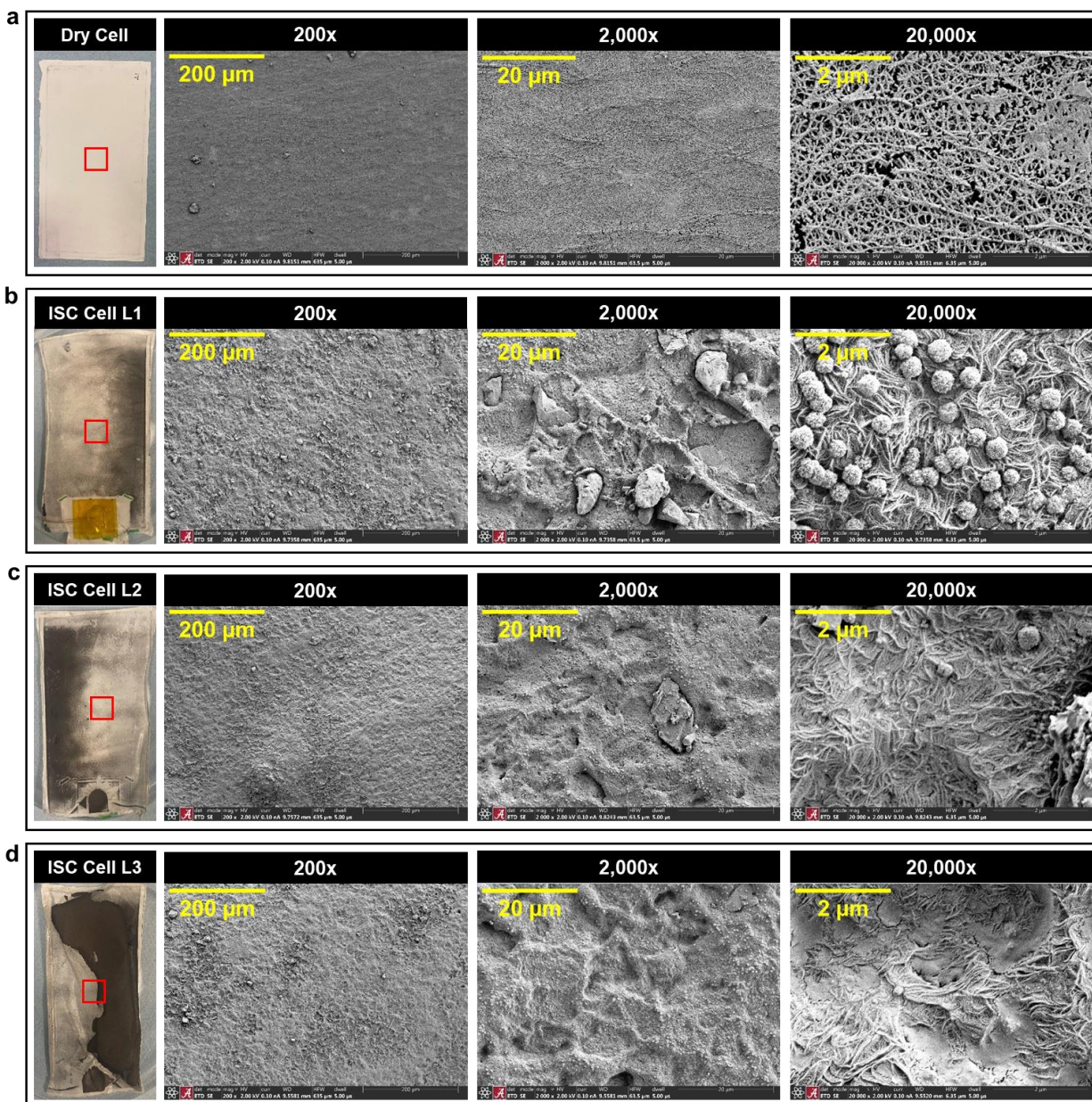
Supplementary Fig. S6 | Discharge performance of the Li-ion cell with Cathode-Cu ISC. a, Before the ISC test. **b,** After the ISC test.



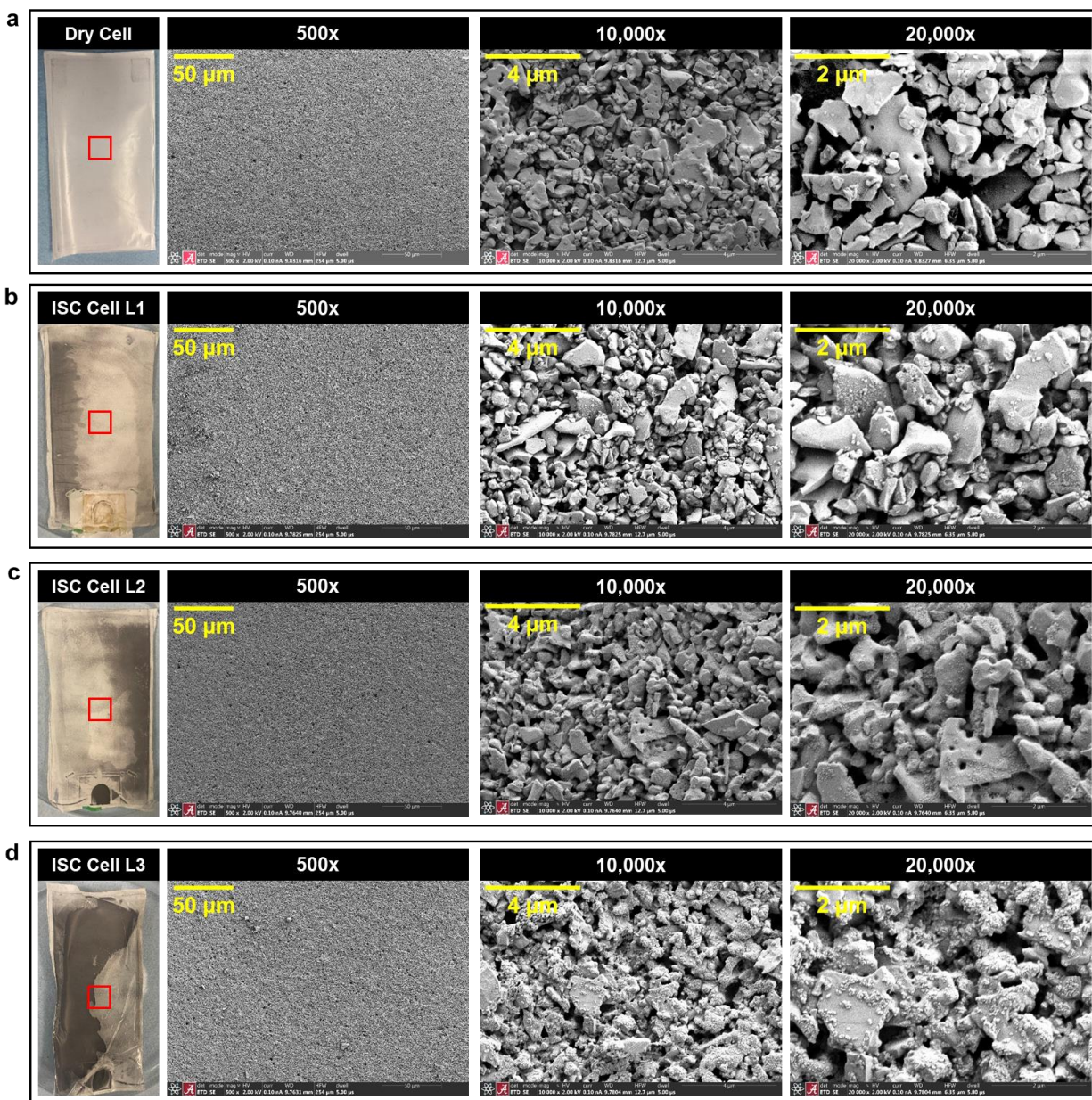
Supplementary Fig. S7 | Comparison of experimental results of Al-Cu ISC cells without and with current modulation. a, ISC temperature. **b,** Cell voltage. **c,** ISC current. **d,** External current. **e,** ISC heat generation rate. **f,** Pictures of cells before and after ISC tests.



Supplementary Fig. S8 | Further analysis of cells with Al-Cu ISC and 120 A current modulation. a-b, Results of temperature, cell voltage and currents during ISC test. **c,** Cell voltage and current during charging and discharging after ISC test.



Supplementary Fig. S9 | SEM images of separators facing anode without ceramic coating.
a, Optical and SEM images (200x, 2,000x, and 20,000x) of a separator from a fresh dry cell. **b-d**, Optical and SEM images (200x, 2,000x, and 20,000x) of different separator layers from Al-Anode ISC cell with current modulation. L1 refers to the innermost layer of separators (this is the separator between the two ISC pads). L2 is the separator directly above L1. L3 is two separators above L1.



Supplementary Fig. S10 | SEM images of separators facing cathode with ceramic coating. **a**, Optical and SEM images (500, 10,000x, and 20,000x) of a separator from a fresh dry cell. **b-d**, Optical and SEM images (500, 10,000x, and 20,000x) of different separator layers from Al-Anode ISC cell with current modulation. L1 refers to the innermost layer of separators (this is the separator between the two ISC pads). L2 is the separator directly above L1. L3 is two separators above L1.

Supplementary Table S1 | Energy balance analysis of Li-ion cells during Al-Anode ISC

Test	E_{ec} (J)	Q_{ec} (J)	Q_{ISC} (J)	Q_{conn} (J)	Q_{ISC} / E_{ec}
Test 1	2933	813	1697	421	58%
Test 2*	4510	1410	2469	636	55%
Test 3*	3458	1030	1899	527	55%
Average	3634	1084	2022	528	56%

*The electrolyte in these cells had different composition (1 M LiPF₆ in EC:EMC = 5:5, MilliporeSigma) from the other cells (1 M LiPF₆ in EC:EMC = 3:7, MTI) due to product discontinuity, which had lower capacities than the cell in Test 1. See more details in Supplementary Table S4.

Supplementary Table S2 | Energy balance analysis Li-ion cells during Al-Anode ISC without or with current modulation

Current Modulation	E_{ec} (J)	Q_{ec} (J)	Q_{ISC} (J)	Q_{conn} (J)	E_{dis} (J)	Q_{ISC} / E_{ec}	T_{max} (°C)
0 A	2933	813	1697	421	0	58%	807
60 A	12758	6972	2387	145	3254	19%	786
120 A max.	13707	10865	679	96	2071	5%	149
External short circuit	15169	13653	206	35	1275	1%	137

Supplementary Table S3 | Energy balance analysis of Al-Cu ISC without or with current modulation

Current Modulation	E_{ec} (J)	Q_{ec} (J)	Q_{ISC} (J)	Q_{conn} (J)	E_{dis} (J)	Q_{ISC} / E_{ec}	T_{max} (°C)
0 A	2336	815	1259	257	0	54%	978
120 A max.	13017	10505	749	112	1748	6%	144

Supplementary Table S4 | Charge capacity, discharge capacity, and coulombic efficiency of cells in formation cycles

	First Cycle			Second Cycle			Third Cycle		
Type	Charge Capacity (Ah)	Discharge Capacity (Ah)	Coulombic Efficiency	Charge Capacity (Ah)	Discharge Capacity (Ah)	Coulombic Efficiency	Charge Capacity (Ah)	Discharge Capacity (Ah)	Coulombic Efficiency
Al-Anode ISC Test 1	4.882	3.844	78.74%	3.937	3.875	98.42%	3.934	3.891	98.89%
Al-Anode ISC Test 2*	4.994	3.529	70.67%	3.603	3.511	97.44%	3.546	3.486	98.31%
Al-Anode ISC Test 3*	5.009	3.430	68.47%	3.510	3.406	97.03%	3.453	3.381	97.91%
Cathode-Anode ISC	4.955	3.986	80.46%	4.064	4.016	98.83%	4.055	4.022	99.18%
Cathode-Cu ISC	4.937	4.069	82.41%	4.164	4.131	99.23%	4.180	4.148	99.23%
Al-Cu ISC	4.904	3.875	79.03%	3.959	3.906	98.66%	3.942	3.904	99.05%
Al-Anode ISC w/ 60A Modulation	4.858	3.891	80.10%	3.983	3.933	98.74%	3.995	3.959	99.12%
Al-Anode ISC w/ 120 A Modulation	4.923	3.868	78.57%	3.971	3.903	98.30%	3.978	3.926	98.69%
Al-Anode ISC w/ ESC Modulation	5.014	4.183	83.43%	4.193	4.100	97.79%	4.123	4.098	99.39%
Al-Cu ISC w/ 120 A Modulation	4.860	3.792	78.03%	3.870	3.814	98.57%	3.879	3.842	99.04%
Standard Deviation	0.059	0.227	4.78%	0.220	0.235	0.68%	0.235	0.248	0.46%

* The electrolyte in these cells had different composition (1 M LiPF₆ in EC:EMC = 5:5, MilliporeSigma) from the other cells (1 M LiPF₆ in EC:EMC = 3:7, MTI) due to product discontinuity, which could cause the lower capacity and coulombic efficiency.